

Capstone Project Report – Sprint 2

Student: Natender

Supervisor: Professor Humera Noor Minhas

Program: MSc. Data Science

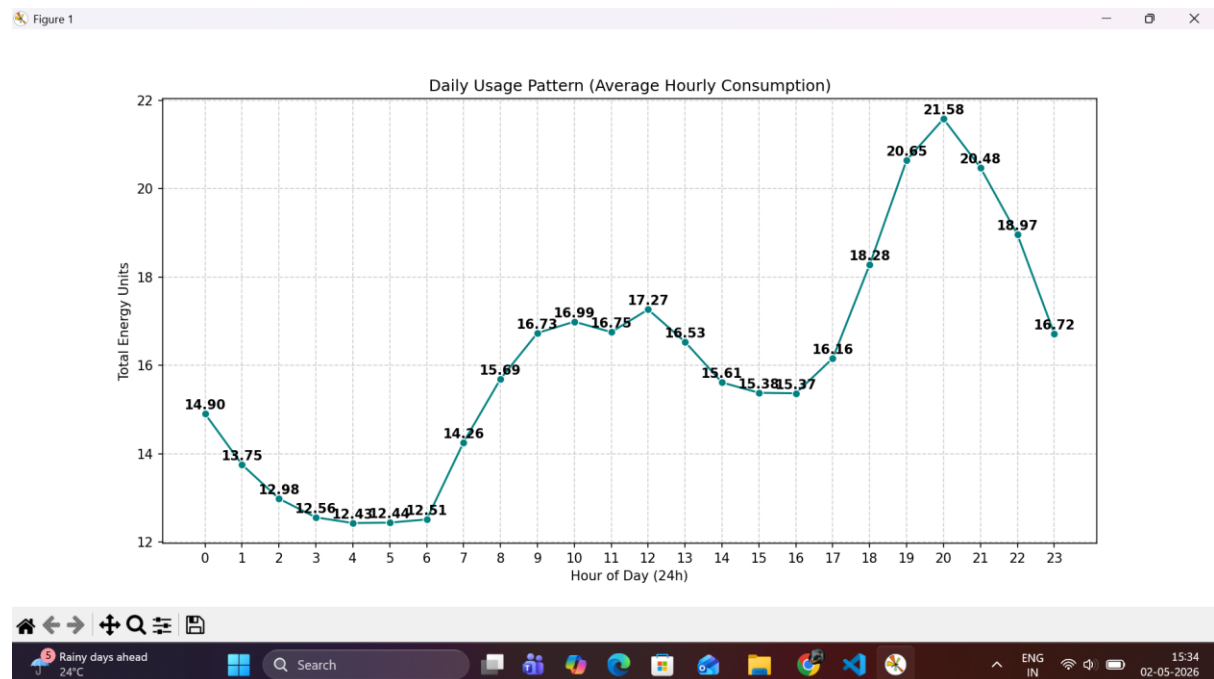
Date: May 2nd 2026

Energy Consumption Analysis Report

1. Daily Usage Patterns

The data reveals a distinct "double-peak" consumption pattern throughout the day, typical of residential or commercial building usage.

- **Morning Peak:** There is a steady rise beginning at **06:00**, peaking around **10:00 to 12:00** with consumption reaching approximately **17.27 units**.
- **Evening Surge (Primary Peak):** The highest usage occurs in the evening, peaking at **20:00** with a maximum of **21.58 units**. This is likely due to lighting, heating/cooling, and appliance use after standard working hours.
- **Baseload Period:** Minimum consumption occurs during the early morning hours (**03:00 to 05:00**), bottoming out at approximately **12.43 units**.

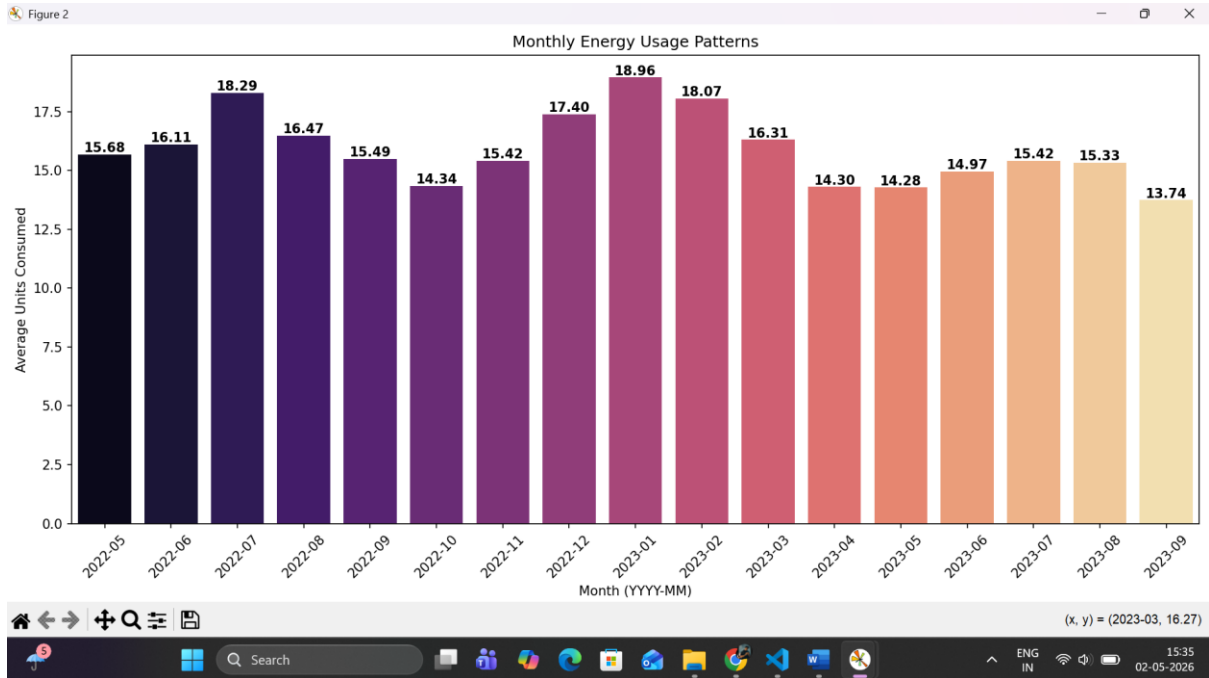


2. Monthly and Seasonal Trends

Energy consumption shows significant seasonal fluctuation over the monitored period (May 2022 – September 2023).

- **Winter Peak:** The highest monthly average consumption was recorded in **January 2023 (18.96 units)**, followed closely by **February (18.07 units)**. This indicates a strong reliance on heating or increased indoor activity during winter months.

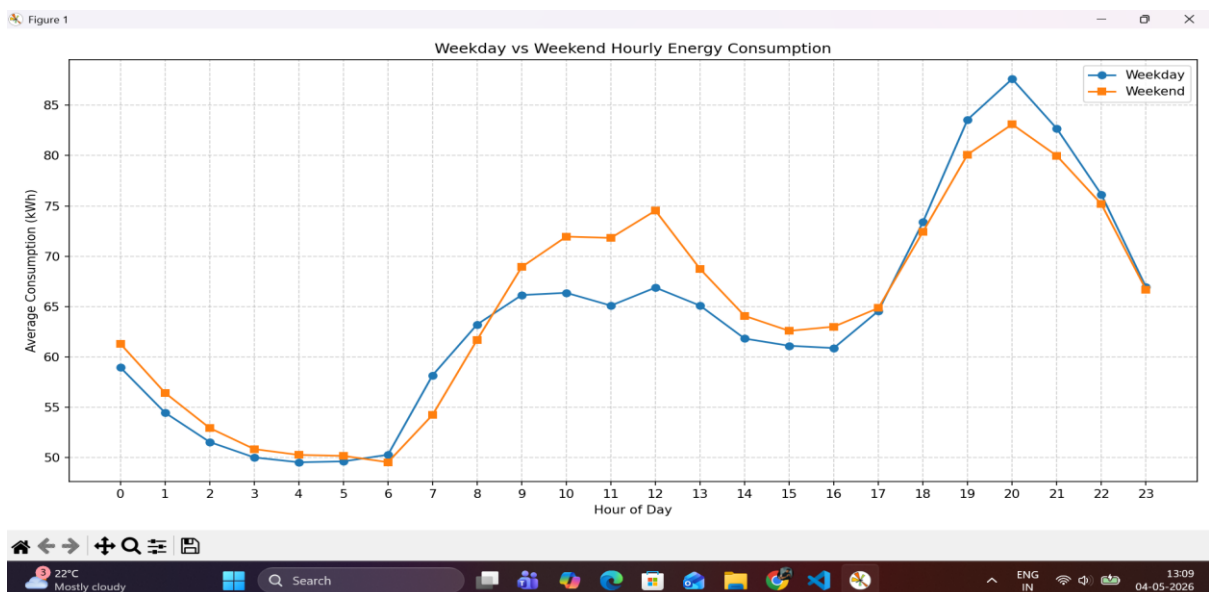
- **Summer Peak:** A secondary peak occurs in **July (18.29 units)**, likely attributed to air conditioning or cooling requirements.
- **Shoulder Seasons:** The lowest usage is observed in **September (13.74 units)** and **April (14.30 units)**, periods where neither significant heating nor cooling is typically required.



3. Weekday vs. Weekend Comparison

Comparing usage types reveals shifts in behavioral timing:

- **Weekends:** Consumption rises earlier in the late morning (around **10:00 - 12:00**) compared to weekdays, suggesting residents are home or active earlier in the day.
- **Weekdays:** The evening peak at **20:00** is significantly higher on weekdays than on weekends. This suggests a more concentrated period of energy-intensive activity when returning from work.



4. Weather Correlations

The correlation analysis shows how environmental factors influence total energy consumption:

- **Relative Humidity (Strongest Negative Correlation):** There is a notable negative correlation (approx. **-0.10**) with humidity. This suggests that as humidity increases, energy consumption tends to decrease, or vice versa.

Possible Reasons

While we usually think of humidity making us feel "sticky," very high relative humidity often occurs during rainy periods or "shoulder seasons" (spring/fall) when the weather is mild.

- ✓ **Rainy Days:** When it rains (100% humidity), outdoor temperatures usually plummet. You don't need the AC, and it's not cold enough yet for heavy heating.
- ✓ **Evaporative Cooling:** In dry climates, people use swamp coolers (evaporative coolers) which use significantly less energy than traditional refrigerant AC units.
- **Wind Speed (Positive Correlation):** Wind speed shows the strongest positive correlation (approx. **0.04**), potentially indicating higher heating needs during windy, colder periods.
- **Temperature and Radiation:** Both Average Temperature and Global Radiation show very weak correlations (near zero), suggesting that while they influence seasonal peaks, they are not the sole drivers of daily consumption fluctuations compared to human behavior.



5. Summary

The energy profile is highly time-dependent, with a primary peak at 20:00 and a secondary peak at midday. Seasonality plays a major role, with January being the most energy-intensive month. To optimize energy costs, demand-response strategies should focus on reducing the 19:00 – 21:00 peak, particularly on weekdays.